

Multimodal-Input-Driven Editable and Controllable 3D Object Manipulation

Project Proposal

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1 Project Goal and Motivation

The goal of this project is to build a model that enables text-driven editing of existing 3D meshes. Given an input 3D model and a textual instruction describing the desired modification, the system should generate a new 3D mesh that reflects the edit while preserving the original structure where appropriate. This approach aims to make 3D editing intuitive and semantically meaningful, reducing the need for manual mesh manipulation and bridging the gap between natural language expression and geometric transformation.

2 Plan of Work

The project begins from the baseline of existing text-to-3D or latent-editing models. We will represent each 3D mesh in a latent space using a pre-trained 3D generative backbone such as DeepSDF or a diffusion-based model. A separate language encoder, similar to CLIP or a transformer-based text model, embeds the textual instruction into the same latent space. The core model will learn to predict a transformation in the latent representation of the input mesh, conditioned on the text embedding, to produce a modified latent code that decodes into the edited 3D mesh. The training process will involve paired examples of original and edited meshes with corresponding text prompts, optimizing for both geometric reconstruction and semantic alignment between the text and the resulting shape.

3 Relation to Existing Research

Recent studies have explored text-guided 3D editing by aligning natural language with 3D generative representations. FocalDreamer introduces a focal-fusion mechanism for localized shape editing from text prompts, demonstrating how language can guide precise modifications without regenerating the entire object [18]. SHAP-EDITOR presents an instruction-conditioned latent editor that performs feed-forward 3D modifications within seconds, avoiding the heavy optimization typical of earlier text-to-3D pipelines [4]. More recently, 3D-LATTE integrates textual conditioning into a 3D diffusion latent space, achieving efficient and semantically aligned editing without retraining or test-time optimization [8]. Together, these approaches reveal the potential of coupling language and geometry, though most operate on implicit neural fields rather than explicit meshes.

Parallel efforts have focused on structure-preserving shape manipulation in latent space. Fine-Grained 3D Vehicle Shape Manipulation via Latent Directions demonstrates interpretable geometric control by learning attribute-specific latent vectors in DeepSDF space [9], while VoxHammer achieves precise and coherent 3D edits through geometry-aware denoising in voxel space [13]. However, these methods lack textual conditioning and remain domain-specific. Our project extends these ideas toward a unified text-driven mesh editing framework, where an input mesh and natural-language instruction jointly drive a latent transformation, producing a new mesh that reflects the requested modification while preserving structural identity.

3.1 Scope Differentiation

Our scope differs in category, modality, control, and evaluation. We target both editing structured data (CAD models, B-rep) and unstructured data (meshes, point clouds). We introduce auxiliary modalities to constrain and edit precisely: text prompts for high-level intent to edit existing 3D models, spatial indicators to localize and bound edits (bounding boxes, segmentation masks) for targeted editing when needed [2, 18], and data indicators for specific edit parameters [12]. Our domain focus is on CAD models (industrial/architectural contexts) where B-reps are standard, starting with available B-rep

datasets and internal conversions while prioritizing datasets that include part/feature semantics to support precise editing and evaluation.

4 Novel Approach and Key Ideas

4.1 Gaps and Limitations We Target

Current methods are insufficient because they struggle with several fundamental challenges that our approach aims to address directly:

Topological Invalidity: Direct generative models often produce invalid geometry (e.g., gaps, non-manifold edges) that is unusable in engineering workflows.

The Precision Gap: Neural networks are inherently approximate, failing to meet the micron-level precision and mathematical constraints (e.g., perfect tangency) required by CAD.

The Complexity Ceiling: Existing models are largely confined to simple sketch-and-extrude operations, unable to generate parts using more complex commands like lofts, sweeps, or revolutions due to data limitations.

Additionally, we address editing both structured and unstructured data, manipulating CAD-based 3D models (B-rep), and introducing auxiliary modalities to constrain/edit precisely. Ideally, text prompts provide high-level intent to edit existing 3D models, while spatial indicators localize and bound edits (bounding boxes, segmentation masks) for targeted editing if needed. Data indicators specify edit parameters. Our goal is to reduce ambiguity and prevent unintended deformations, with a domain focus on CAD models in industrial/architectural contexts.

4.2 Core Insights and Constraints

One of the major constraints is fidelity and controllability, requiring context awareness and fixing the modality. Our core insight is to create a hybrid system that leverages the "best of both worlds." Instead of a single end-to-end model, our approach decouples the problem:

A perception model (e.g., a Vision-Language Model or GNN) will analyze the input (point cloud, sketch, or text) and generate a high-level, symbolic design plan (e.g., "a cylinder joined to a cube with a filleted edge").

This plan is then executed by a symbolic geometric solver. This deterministic component is responsible for enforcing mathematical precision, geometric constraints, and ensuring the final B-Rep is topologically valid.

This division of labor allows the neural network to focus on creative inference and pattern recognition, while offloading the requirements of precision and validity to a system built for logical rigor.

4.3 Risks and Fallback Plans

Risk 1: Integration Complexity

The primary risk is the integration complexity. Creating a seamless and robust interface between the neural planner and the symbolic solver is a significant engineering challenge. The symbolic solver might frequently fail if the neural network's high-level plans are ambiguous or geometrically impossible to resolve.

Risk 2: Multimodal Dataset Availability

One of the most significant challenges lies in assembling or curating a multimodal dataset that includes aligned text, geometry, and semantic labels. Existing datasets primarily consist of unstructured 3D representations (e.g., point clouds, meshes) rather than parametric CAD/B-rep models, limiting geometric precision and editability. Datasets rarely contain text + geometry + semantic label triplets, which are crucial for training cross-modal editing or generation models. Reconstructed CAD/B-rep models are seldom openly available, making data standardization and interoperability difficult.

Baseline/Fallback: Following the strategy in Miao et al. (2024) [10], use a 3D diffusion model for generation, but adopt Miao-style attribute regression in latent space to steer edits. Concretely, train an attribute regressor on the diffusion latent (or intermediate features) and use it to (a) define attribute directions for latent updates or (b) provide attribute-guided conditioning (e.g., via classifier(-free) guidance) during sampling. This ports the 'regression-guided latent editing' idea from Miao et al. to a diffusion backbone.

Risk 3: Editing and Control Risks – Spatial Localization and User Control

Multimodal editing accuracy depends heavily on the model's ability to localize editing regions and interpret ambiguous language inputs. Text prompts such as "extend the top surface slightly" lack geometric specificity, often resulting in over-

editing or unintended deformations. Over-reliance on textual cues without geometric anchors also risks poor spatial alignment.

Fallback: Adopt a bounding-box or segmentation-guided editing workflow inspired by FocalDreamer (Zhang et al., 2023) [18] and SceneFactor (Bokhovkin et al., 2024) [2]. These methods employ region masks, ellipsoid focal regions, or scene factors to confine local edits, ensuring geometric consistency and part-wise control even when text descriptions are vague. This fallback allows the system to leverage 2D images or mask inputs as auxiliary modalities until full text-to-3D alignment is achieved.

5 Evaluation Methodology, Datasets, and Metrics

5.1 Training Objectives

The training framework will likely combine geometric reconstruction with semantic control mechanisms. We anticipate the following loss components:

Geometric Reconstruction Loss: Chamfer Distance (CD): $L_{CD} = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|^2 + \sum_{y \in S_2} \min_{x \in S_1} \|y - x\|^2$. Earth Mover’s Distance (EMD) as an alternative for global shape fidelity. Reference: Fan et al. [5].

Attribute Mapping Loss: Supervised regression between latent codes and semantic attributes: $L_{attr} = \|f_\theta(z) - a\|^2$ where f_θ is a learned regressor, z is latent code, a is target attribute. Reference: Miao et al. [10].

Disentanglement Regularization: Orthogonality constraint between attribute directions: $L_{ortho} = \sum_{i \neq j} |\langle d_i, d_j \rangle|$ where d_i represents learned direction vectors. Alternative approaches may include β -VAE style regularization or contrastive learning. References: Higgins et al. [6]; Chen et al. [3].

Edit Locality Loss (conditional on approach): Spatial masking to constrain edits: $L_{local} = \|M \odot (S_{edit} - S_{orig})\|^2$ where M is a binary spatial mask.

Cross-Modal Alignment (if pursuing multimodal control): CLIP-based text-geometry alignment: $L_{CLIP} = 1 - \cos(E_{text}(p), E_{img}(R(S)))$ where R renders the shape, E are CLIP encoders. Reference: Radford et al. [11].

5.2 Quantitative Evaluation Metrics

Set-Level Generative Metrics: Maximum Mean Discrepancy (MMD) measures distribution distance between generated and reference sets. Coverage (COV) represents the proportion of reference set matched by generated samples. 1-Nearest Neighbor Accuracy (1-NN) tests for mode collapse and overfitting. References: Achlioptas et al. [1]; Yang et al. [16].

Edit-Specific Metrics: Edit Locality measures percentage of geometric change confined to target region via per-vertex displacement analysis. Attribute Alignment uses Spearman correlation between control parameters and measured geometric attributes. Identity Preservation measures Chamfer Distance between original and edited shapes outside edit region.

CAD Validity Metrics (if applicable): Topology Validity (watertight surface check, manifold verification). Novelty/Uniqueness as defined in Willis et al. [14] and Xu et al. [15].

5.3 Qualitative Evaluation

Visual Assessment: Side-by-side comparisons of original and edited geometry, multi-view renderings to assess 3D consistency, interpolation sequences between attribute extremes.

Ablation Studies: We will conduct ablations to isolate the contribution of key components: effect of disentanglement regularizers on attribute independence, impact of locality constraints on edit precision, comparison of different attribute parameterizations.

Comparative Baselines: Potential comparison points include naive latent space arithmetic without learned directions, Miao et al.’s vehicle editor (if replicable) [9], and text-to-3D editing methods (e.g., FocalDreamer, if applicable to our setting) [18].

User Study (time permitting): Task-based evaluation: "Create a shape matching description X." Metrics include control-ability ratings (Likert scale), task completion time, preference rankings. Sample size: target 15-20 participants.

5.4 Datasets

ShapeNet Core (<https://shapenet.org>): 51,300 3D models across 55 categories. Splits: standard 70/15/15 train/val/test per category. Focus categories: cars (7,497 models), chairs (6,778 models). License: Research/non-commercial use.

PartNet (<https://partnet.cs.stanford.edu>): 26,671 3D models with fine-grained part-level annotations. Subset of ShapeNet with hierarchical part segmentation. Essential for part-based editing and locality evaluation. License: MIT.

Objaverse (<https://objaverse.allenai.org>): 800K+ diverse 3D assets (potential scale-up dataset). Would require quality filtering and subset selection. License: Various (CC-BY, CC0, proprietary - requires filtering).

ABC Dataset (<https://deep-geometry.github.io/abc-dataset>): 1M+ CAD models with B-Rep structure. Only relevant if pursuing parametric/CAD editing direction. License: MIT.

Fusion 360 Gallery (<https://github.com/AutodeskAILab/Fusion360GalleryDataset>): 8,625 parametric CAD assemblies with reconstruction sequences. High-quality engineering models. Only relevant for CAD-focused work. License: Non-commercial research.

We anticipate primary reliance on ShapeNet + PartNet, with potential expansion to Objaverse subsets for generalization experiments.

5.5 Success Criteria

The project will be considered successful if:

Minimum: Demonstrate controllable attribute editing on a single category (cars or chairs) with quantitative improvement over naive latent interpolation in at least 2 of 3 metrics (MMD, edit locality, attribute alignment).

Target: Achieve comparable or better controllability than Miao et al. baseline while improving disentanglement (measured via orthogonality of learned directions).

Stretch: Enable cross-modal control (text or sketch) with good CLIP alignment scores, or demonstrate validity-preserving CAD editing with relatively high topology preservation rate.

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